Missing Data Imputation Method for Phylogenetic Trees Based on Singular Value Decomposition and Denoising Autoencoder

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***Abstract*—The expansion of genomic sequencing and advances in genomics have enabled phylogenetic analysis at the whole-genome scale, yet missing data remains a fundamental challenge.This study analyzes the characteristics of phylogenetic matrix imputation and introduces a comprehensive framework for simulating, processing, and analyzing such scenarios, with specialized modules to enhance accuracy. This study introduces SDimp, a novel two-stage imputation method combining low-rank matrix factorization (SVD) for global structure preservation with a denoising autoencoder (DAE) incorporating mask-aware training and recursive updates for nonlinear pattern refinement. Extensive validation on diverse datasets confirms SDimp's superior phylogenetic imputation performance.**

Keywords—Phylogenetic tree, Missing data imputation, Two-stage optimization strategy, Machine learning

I. Introduction

Phylogenetic analysis serves as a fundamental methodology for understanding biodiversity and evolutionary mechanisms. The expansion of genomic sequencing has established genome-scale species tree inference as a core task in comparative genomics and evolutionary biology [1]. Phylogenetic reconstruction methods are primarily categorized into character-based and distance-based approaches [2], with distance-based methods offering computationally efficient solutions particularly suited for large-scale analyses [3]. However, conventional distance-based methodologies necessitate complete matrices without missing entries, while practical datasets frequently contain gaps arising from experimental limitations [4], sampling strategies [5], gene evolutionary dynamics, or technical constraints, compromising phylogenetic accuracy and demanding effective handling strategies [6].

Through systematic analysis of existing imputation methods, this study introduces an optimized phylogenetic imputation framework, integrating specialized modules for simulation, processing, and analysis of phylogenetic missing-data scenarios. Moreover, we present a novel two-stage refinement methodology termed the Singular Value Decomposition and Denoising Autoencoder Imputation Method (SDimp), specifically designed for phylogenetic distance matrices characterized by high missing-data rates. The first stage employs low-rank matrix (SVD) decomposition to exploit evolutionary structures; the second stage utilizes a denoising autoencoder (DAE) with recursive updates and mask-aware training to capture nonlinear patterns for iterative refinement [7]. This synergistic "low-rank initialization & nonlinear optimization" paradigm demonstrates superior performance in both reconstruction accuracy and computational efficiency.

II. System and Methods

Current research on missing value handling in phylogenetic matrices remains limited. These research efforts primarily rely on two approaches: the direct approach and the indirect approach [8]. Direct methods like the triangles method [9], least-squares (LS) method named MW-modified [10], and adapted neighbor-joining (NJ) operate directly on partial matrices but face combinatorial constraints restricting practical application [11]. Indirect approaches employ a two-stage imputation-then-construction process, though conventional phylogenetic completion substitutes missing inter-taxon distances with arithmetic means or medians of available measurements can introduce systematic biases affecting phylogenetic accuracy [12, 13, 14]. Early methodological developments encompassed whole-genome distance approaches incorporating risk analysis [15], alongside composite vector techniques for genomic data [16].

Recent advances in phylogenetic reconstruction from incomplete data favor hybrid strategies merging direct and indirect approaches. LASSO heuristically handles incomplete matrices but relies on the restrictive molecular clock assumption [17]. DAMBE employs multivariate least squares with accuracy declining under high missing rates [18]. Modern machine learning techniques include parameter-intensive MF/AE methods using regularized matrix factorization and autoencoders [19], and PhyloMissForest applying phylogenetically-guided random forests [20]. Collectively, MF, AE, and PhyloMissForest (PhF) constitute cutting-edge indirect imputation strategies, each offering distinct advantages across varying missing-data contexts while preserving biological validity in evolutionary inference.

III. Phylogenetic Tree Imputation Framework

Our phylogenetic imputation framework comprises four methodical phases: 1) missing data simulation through randomized deletion, 2) advanced gap imputation, 3) tree reconstruction via NJ algorithm [21], and 4) topological validation using Robinson-Foulds metrics [22]. The core

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imputation phase integrates three specialized modules-Initial Value Selection, Optimal Matrix Strategy, and Matrix Missing Rate Reduction-collectively engineered to address phylogenetic completion constraints while ensuring evolutionary plausibility.

A. Initial Value Selection

Prior to imputation, missing matrix values require phylogenetically constrained initialization. For a complete matrix V with maximum observed distance d_{max}, any missing distance d_{ij} between leaves i and j must satisfy $0 < d_{ij} < d_{max}$ while maintaining consistency with distances to other leaves $\partial \in N$ ($\partial \neq$ i, j). When a pairwise leaf distance is missing in an n-leaf phylogenetic tree, it can be estimated using available distances from other leaves within the corresponding clade through the following computational principle:

$$\Delta = \max ||d_{i\partial} - d_{j\partial}|| \qquad \partial \in N \,\&\, \partial \neq i,\ j \tag{1}$$

$$\overline{d} = \frac{\sum_{\partial \in N}(d_{i\partial} + d_{j\partial})}{2(n-2)} \qquad \partial \neq i,\ j \tag{2}$$

The method constrains the initial guess values such that the distance d_{ij} between leaves i and j must fall within the interval $[0, \overline{d} + \Delta]$.

B. Optimal Matrix Strategy

To address potential violations of phylogenetic matrix properties following algorithmic iterations, we introduce a strategic approach in our framework. Each iteration generates three distinct candidates: the upper triangular matrix, lower triangular matrix, and their arithmetic mean. Optimal matrix selection is achieved through a least-squares phylogenetic fit criterion, thereby ensuring maximal preservation of evolutionary relationships at each imputation stage.

$$S = \sum_{i,j} (\delta_{ij} - \theta_{ij})^2 \tag{3}$$

For each candidate matrix V, a phylogenetic tree T is constructed. The squared difference between matrices is then calculated using (3), where δ_{ij} represents the distance between i and j in matrix V, and θ_{ij} denotes the corresponding distance in the phylogenetic tree T derived from V. The matrix yielding the smallest S value is selected as the optimal output. This approach not only ensures comprehensive exploration of all viable imputed matrices, but also guarantees the selection of the matrix that best conforms to phylogenetic tree properties.

C. Matrix Missing Rate Reduction

Genomic missing data demonstrate significant heterogeneity in phylogenetic matrices, where certain genomes exhibit substantially higher missing rates that challenge imputation accuracy [23]. Empirical evidence establishes that reliable imputation performance is typically confined to a specific High-Confidence Range [0, k]. To address matrices exceeding this threshold, we implement a strategic decomposition protocol: for an m×m matrix V with missing rate w>k, genomes with severe data gaps are temporarily excluded, generating an n×n matrix V_1 with normalized missing rate within [0, k]. Following V_1's imputation, excluded genomes are reintegrated to form matrix V_2, which undergoes final completion. Although introducing moderate computational overhead, this approach significantly enhances reconstruction accuracy as experimentally verified.

D. Results

To validate framework efficacy, we conducted comparative experiments assessing AE and MF methods using three phylogenetic datasets: 9 baculovirus, 37 Auricularia amino acid, and 55 green plant DNA sequences. Missing rates from 10-60% (10% increments) were tested with five matrices per rate, executing 10 repetitions per configuration for statistical reliability. Topological accuracy was measured via normalized Robinson-Foulds scores (0-1 range, lower values indicating better congruence with reference trees). NMF and NAE represent implementations within the imputation framework, with boldface indicating superior results. All experiments were performed on an Intel i5-12600KF/32GB DDR4 system using Python 3.11.4.

Table Ⅰ validates our framework's robust performance across datasets, demonstrating particularly significant enhancement in AE imputation for the medium-scale Auricularia amino acid dataset (37×37). While maintaining secondary improvements on small-scale data, the framework shows diminished efficacy on 55×55 matrices with negligible quality gain at 40% missing rates versus baselines. These results collectively establish the framework's capacity to substantially improve imputation quality for small-to-medium phylogenetic datasets, achieving optimal performance at intermediate scales.

TABLE I. IMPUTATION FRAMEWORK EXPERIMENTAL DATA

Taxa	Entries	Missing	NRF Rate			
			MF	*NMF*	*AE*	*NAE*
9	36	10%	0.155	**0.133**	0.344	**0.288**
		20%	0.344	**0.288**	0.366	**0.312**
		30%	0.366	**0.266**	0.392	**0.222**
		40%	0.410	**0.320**	0.494	**0.222**
		50%	0.586	**0.488**	0.532	**0.334**
		60%	0.677	**0.522**	0.561	**0.455**
37	666	10%	0.203	**0.191**	0.356	**0.185**
		20%	0.235	**0.190**	0.431	**0.232**
		30%	0.309	**0.230**	0.553	**0.168**
		40%	0.391	**0.326**	0.559	**0.283**
		50%	0.445	**0.320**	0.680	**0.355**
		60%	0.635	**0.443**	0.816	**0.486**
55	1485	10%	0.205	**0.138**	0.239	**0.181**
		20%	0.238	**0.158**	0.238	**0.198**
		30%	0.267	**0.257**	0.387	**0.247**
		40%	0.258	0.258	**0.336**	0.337
		50%	0.357	**0.306**	0.357	**0.258**
		60%	0.486	**0.396**	0.476	**0.406**

IV. SDIMP PHYLOGENETIC TREE IMPUTATION METHOD

This study presents an integrated phylogenetic imputation method combining SVD-based low-rank completion with enhanced deep learning. The two-stage SDimp framework

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employs SVD decomposition for initial structural recovery, followed by an enhanced DAE. Extensive validation confirms superior accuracy under high missing-rate scenarios.

A. SDimp Workflow

SDimp implementation progresses through three sequential phases: initial matrix construction encodes incomplete data as $V_{missing}$ with NaN markers, generating V_0 via mean imputation; preliminary completion performs SVD decomposition on V_0 yielding V_1; recursive optimization stage employs a DAE processing concatenated inputs [V_1, M]-where Boolean mask M distinguishes observed from missing values-through recursive updates with decaying weights to produce final imputed matrix V_2. Fig.1 illustrates the comprehensive architecture of the SDimp framework.

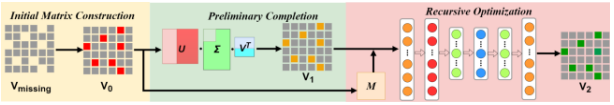

Fig. 1. SDimp framework

B. SVD-Based Preliminary Completion

Phylogenetic distance matrices typically exhibit low-rank or approximately low-rank properties, concentrating evolutionary information within few principal components [24]. SVD enables effective low-rank approximation through dominant pattern extraction, exhibiting superior noise robustness over local methods [25]. This global approximation facilitates reliable estimation of unknown leaf-pair distances through available inter-leaf measurements. The preliminary imputation employs nuclear norm-regularized completion via SoftImpute, executing: iterative SVD with soft-thresholded singular values for progressive approximation, exclusive missing-entry updates preserving data integrity, and optimal output selection ensuring matrix symmetry.

Matrix V_0 undergoes SVD decomposition with singular value shrinkage enforcing low-rank constraints through minor value truncation. The regularization parameter λ governs rank reduction intensity, where elevated values promote sparsity at potential evolutionary signal cost. Dynamic λ adjustment via (4) mitigates overfitting prior to reconstruction, maintaining critical balance between dimensionality reduction and phylogenetic information preservation. Convergence is achieved at $|X_{k+1} - X_k| < 10^{-6}$.

$$\lambda = 0.1 + 0.9 * W \tag{4}$$

While SVD-based methods provide initial imputation through low-rank linear assumptions, they face inherent limitations: inability to capture nonlinear evolutionary relationships and predominant focus on global patterns at the potential expense of local phylogenetic structures.

C. DAE-Based Recursive Optimization

To address these limitations, we developed an enhanced DAE that refines initial SVD-based imputation through nonlinear pattern learning. This architecture captures complex data distributions while correcting linear biases, establishing a synergistic "initial completion + refinement" paradigm. The improved DAE employs mask-aware input by concatenating data matrices with missing-value indicators into 2n-dimensional inputs, processed through dynamically shrinking encoder layers with dropout regularization. The optimization executes recursive updates through: noise injection for robustness enhancement, reconstruction error minimization at observed positions, exponentially-weighted blending of predicted and current values, and biological constraint enforcement. A dual termination criterion—combining maximum iterations and minimum error thresholds—ensures progressive bias correction.

$$X' = (1 - w)X + wp \qquad w = 0.8^T \tag{5}$$

We implements a data-mask concatenation architecture that utilizes mask matrix M to identify imputation targets within preliminary matrix V_1. The recursive phase integrates three core operations: noise-injected training, dynamic weight decay, and optimal matrix selection, governed by dual termination criteria (maximum iterations T and error threshold 10^{-6}) to generate final matrix V_2. This structured approach promotes generalized feature learning by reducing dependency on imputed values while adaptively modulating parameters through dynamic weight decay. The method stabilizes convergence through parameter drift constraints while maintaining optimization integrity via weight updates in (5), crucially mitigating overfitting in high-dimensional phylogenetic data with limited samples.

D. Results

Our experimental setup maintains consistency with previously described datasets and evaluation metrics. Table III presents a comprehensive comparison of SDimp against three leading machine learning imputation methods—MF, AE, and PhF—alongside two classical approaches, DAMBE and LASSO, to validate its effectiveness. For computational efficiency assessment, we performed comparative runtime analysis across varying dataset scales. Table Ⅱ documents the average execution times recorded for MF, AE, PhF, and the proposed SDimp method. The LASSO method demonstrated substantial time instability, while DAMBE required prohibitive processing time for larger matrices or elevated missing rates, necessitating their exclusion from final comparison. Boldface denotes optimal metric performance, underlined values indicate secondary results, and N/A signifies incomplete matrix decomposition within 48 hours.

Table III demonstrates SDimp's robust performance across datasets, achieving optimal or near-optimal accuracy under most missing-rate conditions. While showing limited advantage on the 9-genome dataset (trailing PhF by 11.5% and MF by 15.5% at 10% missing rate), SDimp exhibits dimensional scalability: at 20% missing rate, NRF scores decrease from 28.9% (9×9) to 18.4% (37×37, 36.3% improvement) and further to 9.9% (55×55, 65.7% improvement over 9×9). The method consistently ranks top-tier for ≥ 37-genome datasets, maintaining computational efficiency despite 66.7% longer runtime than MF on 37×37 data. These results validate SDimp's capacity to: 1) handle phylogenetic missing data across scales, 2) significantly enhance accuracy, and 3) preserve efficiency while demonstrating dimensional scalability.

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TABLE II. MEAN EXECUTION TIMES

Taxa	MF	AE	PhF	SDimp
9	**18s**	25s	20s	30s
37	17min	**37s**	20min	75s
55	53min	2min	25min	**112s**

TABLE III. IMPUTATION METHOD EXPERIMENTAL DATA

Taxa	Entries	Missing	NRF Rate					
			MF	*AE*	*DAMBE*	*LASSO*	*PhF*	*SDimp*
9	36	10%	0.155	0.344	0.292	0.247	**0.115**	0.216
		20%	0.344	0.366	0.386	0.259	**0.158**	0.289
		30%	0.366	0.392	0.515	0.415	0.176	0.242
		40%	0.410	0.494	0.641	0.438	0.217	**0.213**
		50%	0.586	0.532	N/A	0.478	0.337	**0.333**
		60%	0.677	0.561	N/A	0.525	**0.355**	0.405
37	666	10%	0.203	0.356	0.359	0.254	**0.105**	0.185
		20%	0.235	0.431	0.445	0.355	0.191	**0.184**
		30%	0.309	0.553	0.658	0.480	0.176	**0.107**
		40%	0.391	0.559	N/A	0.583	**0.213**	0.246
		50%	0.445	0.680	N/A	0.633	0.281	**0.261**
		60%	0.635	0.816	N/A	0.689	**0.302**	0.366
55	1485	10%	0.205	0.239	N/A	0.216	0.092	**0.082**
		20%	0.238	0.238	N/A	0.249	0.111	**0.099**
		30%	0.267	0.387	N/A	0.294	**0.146**	0.158
		40%	0.258	0.336	N/A	0.343	**0.203**	0.207
		50%	0.357	0.357	N/A	0.472	0.242	**0.178**
		60%	0.486	0.476	N/A	0.581	**0.322**	0.356

V. CONCLUSION

This study introduces a customizable phylogenetic imputation framework and proposes SDimp—a novel method integrating SVD and DAE. Experimental validation demonstrates robust performance across diverse datasets, with particular efficacy on medium-to-large scale phylogenetic data.

By synergistically combining SVD's structural preservation with DAE's nonlinear modeling, SDimp achieves balanced capture of global and local data characteristics, substantially advancing imputation accuracy. The framework further provides extensible optimization principles transferable to related bioinformatics domains. Future work will focus on DAE architecture optimization, dynamic weight decay strategies, extension to broader bioinformatics missing-data problems, and incorporation of formal methods for enhanced algorithmic reliability and reproducibility.

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